CS240: Database Systems

Lecture 01:

Introduction to Database systems

Overview

- **Database & DBMS**
- Data Models
- Application & History

What is a Database?

- A collection of data, typically describing the activities of one or more related organizations.
- Models real-world enterprise
  - Entities
    - e.g. students, professors, courses, classrooms
  - Relationships between entities
    - e.g. student's enrollment in courses, professor teaching courses, and use of room for courses.

What is a Database?

- Databases play a critical role in almost all areas
  - **Banking:** all transactions
  - **Airline:** reservation, schedules
  - **Universities:** registration, grades
  - **Sales:** customers, products, purchases
  - **Manufacturing:** production, inventory, orders, supply chain
  - **Human resources:** employee records, salaries, tax deductions

What is a Database?

- A database can be of **any size** and of **varying complexity**.
  - For example, the list of names and address of friends
  - The book catalog of a large library may contain half a million records
  - A database of much greater size and complexity is maintained by the government to keep track of the tax information filed by taxpayers.

What is a Database?

Student Name	ID	Age	Gender	Entrance Year	Grade
Chan Mei Yee	A34455	20	F	1998	A
Lee Wai Man	C23444	19	M	1999	B
Wong Wing Nam	C73334	19	M	2000	C

- A **schema** is the ***definition*** of a database. It defines the ***meaning*** of data.
- An **instance** of a database is ***the collection of data*** in the database at a particular point of time (snapshot).
- For example, in the above, the schema is "*Student Name, ID, Age, Gender, Entrance Year, Grade*". The remaining rows in the table make up an instance of the database.

What is a DBMS?

- **DBMS – Database Management System**
- A DBMS is a collection of ***software programs*** to enable users to ***create***, ***maintain*** and ***utilize*** a ***database***.

What is a DBMS?

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- DBMS
  - Insert records
  - Delete records
  - Update records
  - Query records

What is a DBMS?

- Commercial DBMS

Company	Product
Oracle	Oracle 8i, 9i, 10g
IBM	DB2, Universal Server
Microsoft	Access, SQL Server
Sybase	Adaptive Server
Informix	Dynamic Server

Can we do without a DBMS?
File System

- Sure! Start by storing the data in files:

student.txt   course.txt   teacher.txt

- Now write C or Java programs to implement specific tasks…
  - Add new students, courses and teachers

Why do we need a DBMS?

- Drawback of file system:

1. Data redundancy and inconsistency
   - E.g., consider a bank application
     - *Address of a customer* in
       - the file of “saving-accounts” and
       - the file of “checking-accounts”

A good design of DBMS can avoid data redundancy and inconsistency.

Why do we need a DBMS?

2. Difficulty in accessing data

- Need to write a *new program* to carry out each *new task*

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It is easy to obtain data with DBMS

Why do we need a DBMS?

3. Integrity problems
   - E.g., consider a bank application
     - The balance *cannot be* below $1000
     - The day of a month *cannot exceed* 31

DBMS can check the integrity automatically

Why do we need a DBMS?

4. Atomicity problems
   - E.g., consider a bank application
     - We want to transfer $100 from account A to account B
     - Steps:
       - **Step 1:** We deduct $100 from account A
       - **Step 2:** Then, we increment $100 in account B
     - If the system *crashes* at Step 1, then Step 2 cannot be executed

DBMS makes sure that Step 1 and Step 2 can be executed together even with a crash (We call the execution is **atomic**.)

Why do we need a DBMS?

5. Concurrent Access by multiple users

- Uncontrolled concurrent accesses can lead to inconsistencies
- E.g., consider a bank application
  - There is an account shared by 2 customers A and B
  - Customers A and B withdraw $1000 concurrently

A	B
Read 5000	
	Read 5000
5000 - 1000	
	5000 - 1000
Write 4000	
	Write 4000

DBMS makes sure that the concurrent access cannot lead to this problem

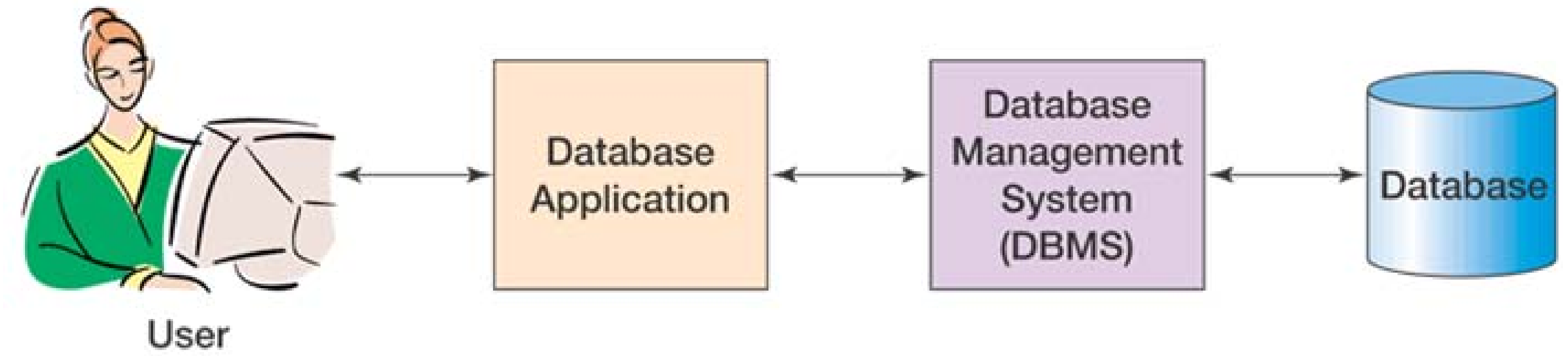
Why do we need a DBMS?

6. Security Problems
   - E.g., consider a bank application
     - We do not want system programmers to have permissions to read some data (e.g., Andy Lau's saving account and Joey Yung's saving account)
   - Need a lot of effort to re-write a program for this permission system

DBMS can enforce that different users have different permissions to access different parts of the data

Advantages of DMBS

- With the use of DBMS, we have the following advantages
  - Data independence
  - Efficient data access
  - Data integrity and security
  - Data administration
  - Concurrent access and crash recovery
- **Overall:** Reduced application development time



Advantages of DMBS

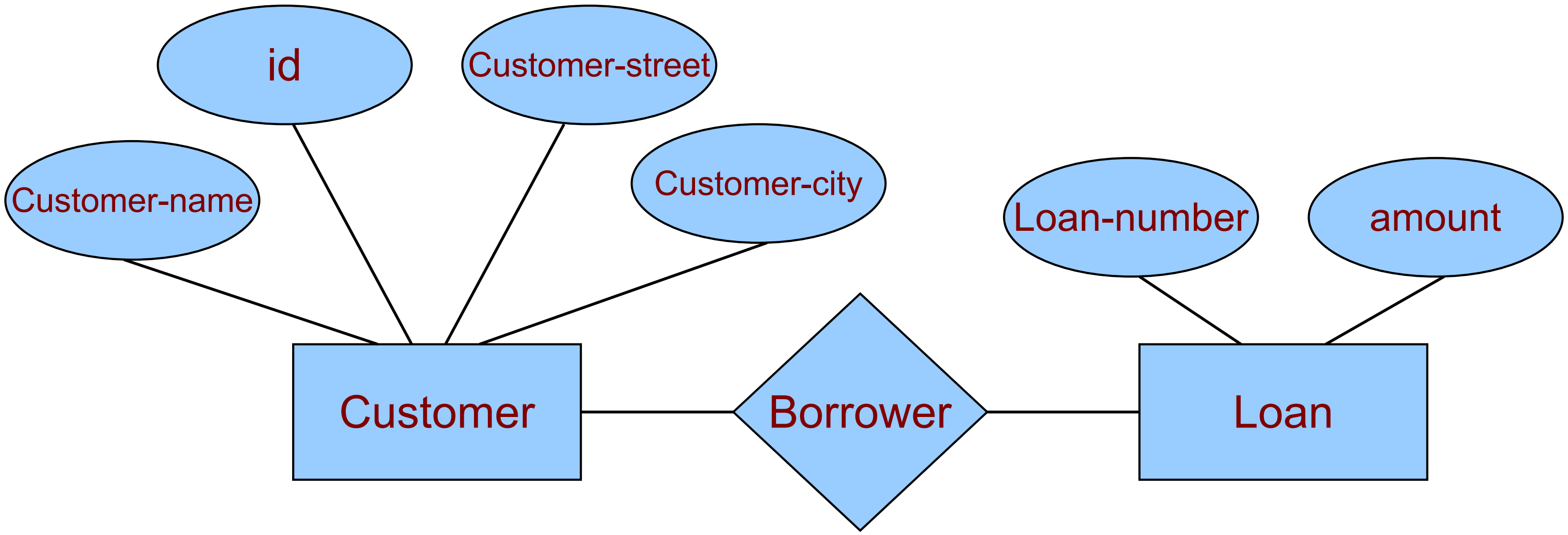
- Data Independence
  - Applications insulated from how data is structured and stored.
  - Logical data independence: Protection from changes in logical structure of data.
  - Physical data independence: Protection from changes in physical structure of data.

Overview

- Database & DBMS
- **Data Models**
- Application & History

Data Models

- A data model is a collection of tools or concepts for describing data, the meaning of data, data relationships and data constraints.
  - **Entity-Relationship Model (ER Model)**



Data Models

- **Relational Model**

Customer-Name	ID	customer-street	customer-city

  - Main concept: **relation**, basically a table with rows and columns. A column is also called a **field** or **attribute**
  - Other models such as the Network Model, Hierarchical Model, Objected-relational Model

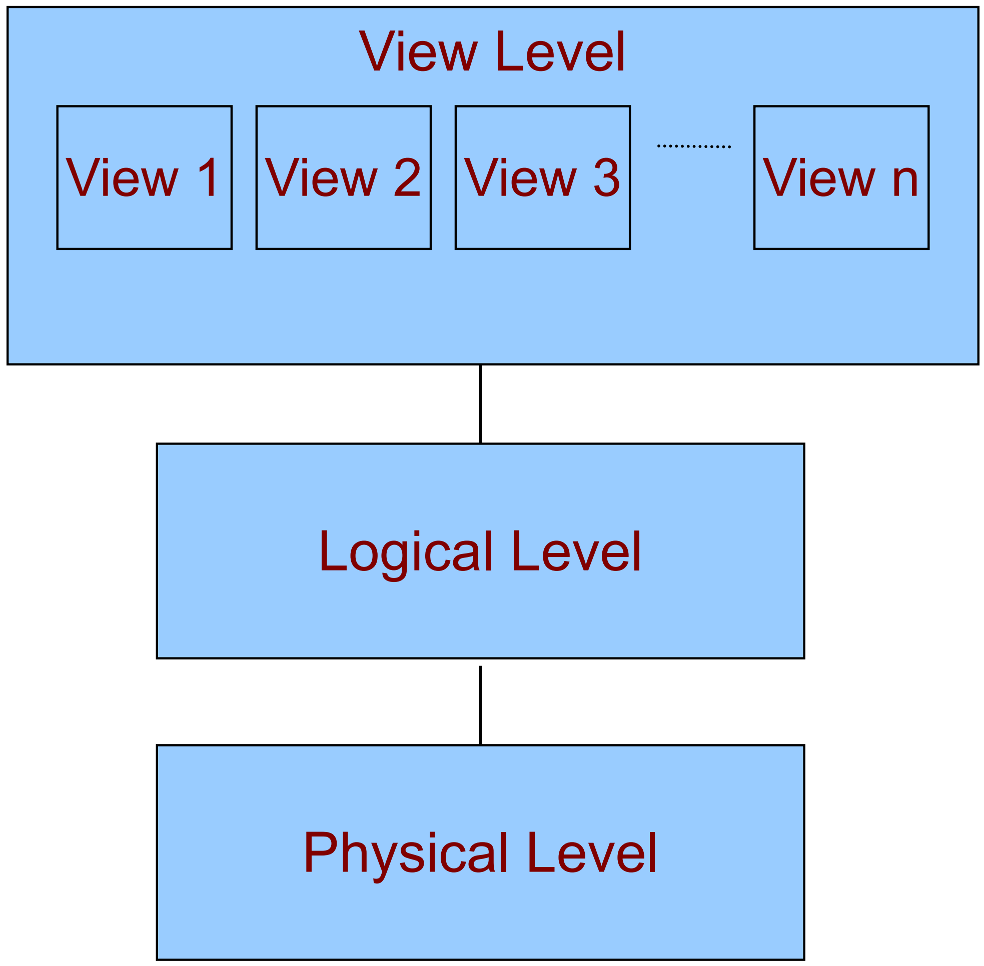
We will focus on the dominant ***Relational model***.

A description of data in terms of a data model is called a **schema**.

Data Abstraction

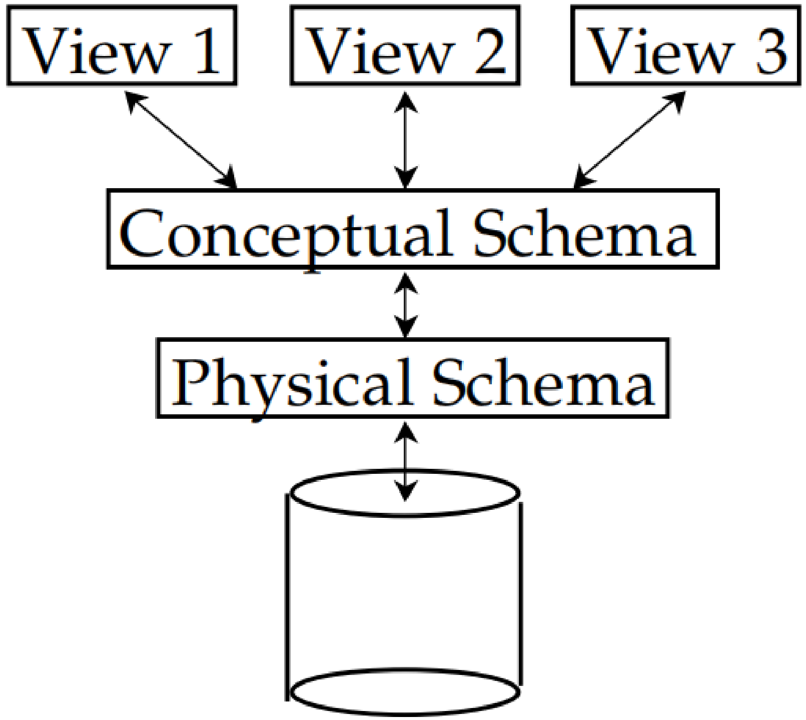
- Hide certain details of how data is stored and maintained
  - **Physical level:** how and where data are actually stored, low level data structures are specified at this level
  - **Conceptual level (logical level):** describes what data should be stored in the database, and relationship and semantics of the data
  - **View level:** Relevant partial view of the database to be particular type of users

Data Abstraction



Levels of Abstraction

- Many views, single conceptual (logical) schema and physical schema.
  - Views (External schema) describe how users see the data.
  - Conceptual schema defines logical structure
  - Physical schema describes the files and indexes used.



Example: University Database

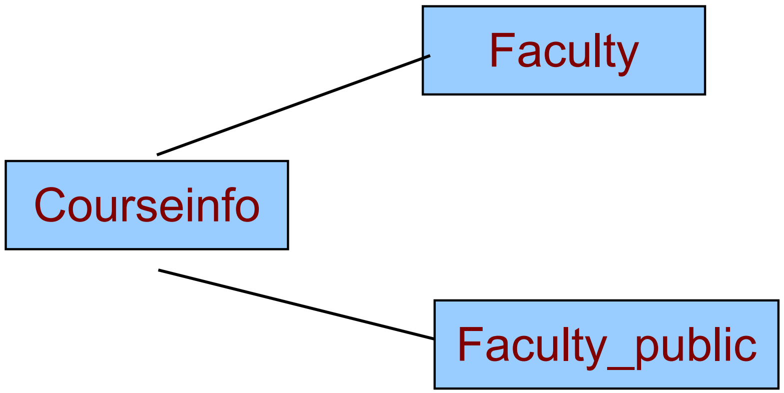
- Conceptual schema:
  - Students(sid: string, name: string, age: integer, gpa:real)
  - Courses(cid: string, cname:string, credits:integer)
  - Faculty(fid: string, fname:string, sal:real)
  - Enrolled(sid:string, cid:string, grade:string)
- Physical schema:
  - Relations stored as unordered files. Index on first column of Students.
- External Schema (View):
  - Courseinfo(cid:string, fname:string, enrollment:integer)

Example: University Database

- Data Independence

  Suppose that the Faculty relation in the university database is replaced by the following two relations:

  - Faculty_public(fid: string, fname:string, office:integer)
  - Faculty_private(fid: string, sal:real)



Database Languages

- **Data Definition Language (DDL)**
  - A language that specifies data schemas
- **Data Manipulation Language (DML)**
  - A language to facilitate the retrieval, update of data in the database
- For retrieval, we query the database with the **query language**, which is part of the DML

What languages does the computer speak?

- Start with SQL DDL to *create tables*:

```
CREATE TABLE Students (
        Name CHAR(30),
        SSN CHAR(9) PRIMARY KEY NOT NULL,
        Category CHAR(20),
);
```

- Continue with SQL to *populate tables:*

```
INSERT INTO Students
VALUES('Hillary', '123456789', 'undergraduate');
```

Querying: Structured Query Language

- Find all the students who have taken CS101:

```
SELECT SSN
FROM   Takes
WHERE  CID='CS101';
```

- Find all the students who CS101 *previously*:

```
SELECT SSN
FROM   Takes
WHERE  CID='CS101' AND Semester='2010';
```

- Find the students' *names*:

```
SELECT Name
FROM   Students, Takes
WHERE  Students.SSN = Takes.SSN AND
       CID='CS101' AND Semester='2010';
```

Structure of a DBMS

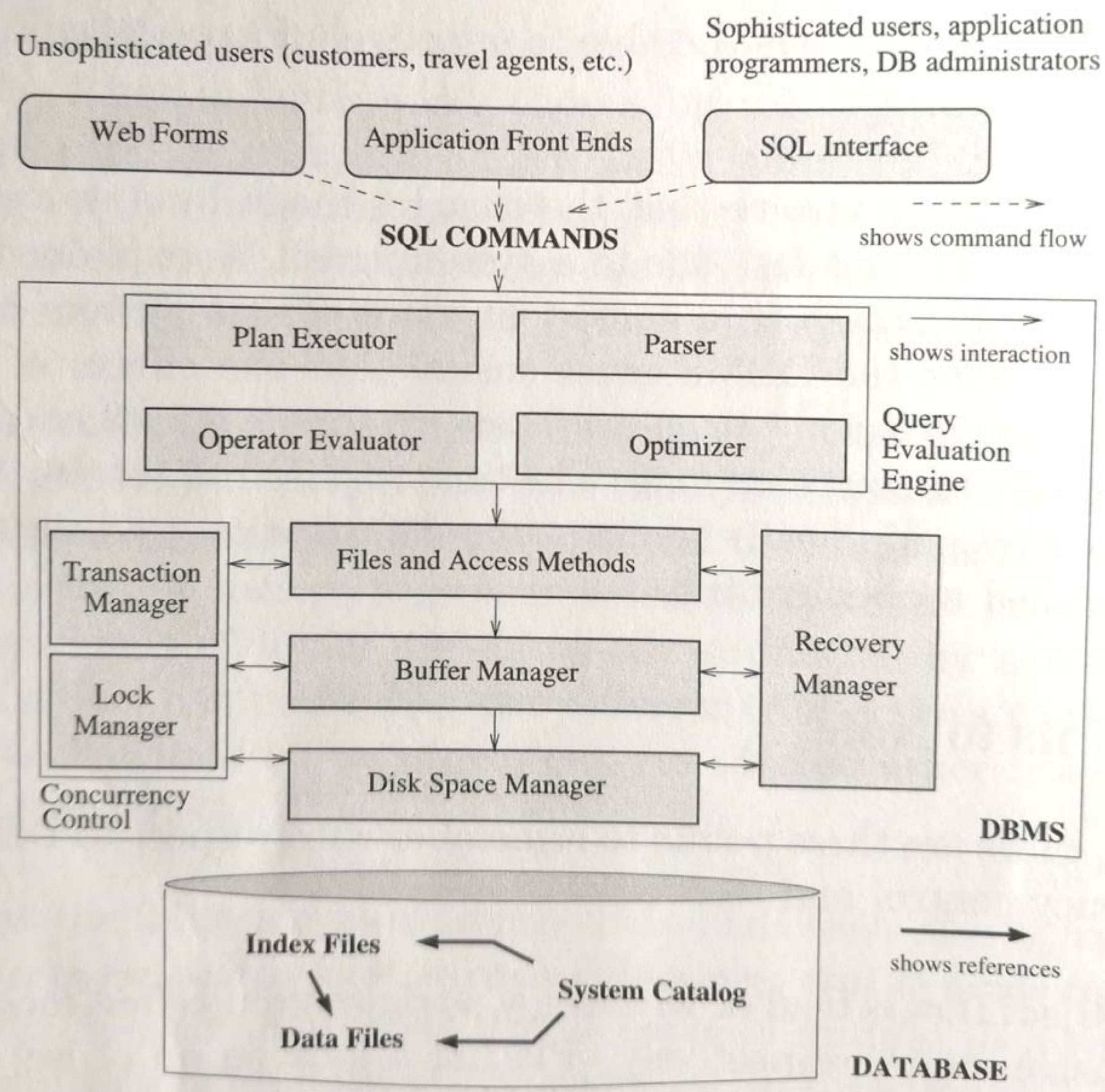


Figure 1.3 Architecture of a DBMS

Overview

- Database & Database Systems
- Data Models
- **Application & History**

People who Deal With Databases

- **Database Administrator (DBA):** Person(s) who has central control over the database and is responsible for the following tasks:
  - Schema definition/modification
  - Storage structure definition/modification
  - Authorization of data access
  - Integrity constraint specification
  - Monitoring performance
  - Responding to changes in requirements

People who Deal With Databases

- **Application Programmers**
  - Embed DML calls in program written in a host language (e.g., Cobol, C, Java). (DML stands for data manipulation language)
  - e.g., programs that generates payroll checks, transfer funds between accounts
- **Sophisticated Users**
  - Form request in database query language
- **Naive users**
  - Invokes one of the permanent application programs that have been written previously
  - e.g. transfer – transfer fund between accounts

Types of Databases and Database Applications

- Traditional Applications:
  - Numeric and Textual Databases
- More Recent Applications:
  - Multimedia Databases
  - Geographic Information Systems (GIS)
  - Data Warehouses
  - Real-time and Active Databases
  - Many other applications

History

- The first DBMS was designed by Bachman at GE in early 1960s
- In 1970 Codd at IBM proposed a new data representation framework called the **relational data model**.
- The SQL query for relational databases, developed as part of IBM's System R project, was standardized in the late 1980s.
- The current standard, SQL-92, was adopted by ANSI (American National Standards Institute) and ISO (International Standards Organization).

History

- In late 1980s and 1990s, several vendors (e.g., IBM's DB2, Oracle 8) have extended their systems with the ability to store new data types such as images and text.
- Specialized systems developed for **data warehouses**, consolidating data from several databases.
- Entering the Internet Age, a new markup language **XML** is proposed for data access through a Web browser.
- As more and more data are collected, companies are also interested to **mine** useful information from their data.

Summary

- What is a Database?
- What is a DBMS?
- Why do we need a DBMS?
- Data models, data abstraction and Data Independence
- DBMS languages and people with DBMS
- Read Chapter 1